

ROLL NUMBER

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Test Series - NEET**PART TEST - XI / 02****NEET(UG)-2026***Do not open this Test Booklet until you are asked to do so.*

Time Allowed: 3 Hours 20 Minutes	Maximum Marks: 720	Number of Questions: 180
Physics (Sec A: 35, Sec B: 10)	Chemistry (Sec A: 35, Sec B: 10)	Biology (Botany: 45, Zoology: 45)

IMPORTANT INSTRUCTIONS / .9\$M5*B0M# (?0M&G6

1. The test consists of 180 questions divided into four subjects: Physics, Chemistry, Botany, and Zoology.
2. Each correct answer carries 4 marks (+4). For each incorrect response, one mark (-1) will be deducted from the total score.
3. Unanswered/unattempted questions will be given no mark (0).
4. Use Blue/Black Ball Point Pen only for writing particulars on this page and marking responses on the OMR Answer Sheet.
5. The test is strictly of 3 hours 20 minutes duration (200 minutes).
6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the OMR Answer Sheet to the invigilator before leaving the Room/Hall.
8. Candidates are allowed to take away this Test Booklet with them after conclusion of the examination.
9. The candidate should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet.
10. Electronic devices, mobile phones, calculators, and slide rules are strictly prohibited in the Examination Hall.

CANDIDATE DECLARATION / *0@ M7>0M% @ K7#>

I have read all the instructions given above and adhere to the guidelines. I hereby declare that the particulars filled in by me are true and complete.

Candidate's Name: _____

Candidate's Signature: _____

Invigilator's Signature: _____

PHYSICS / - L\$? @

1. Two identical springs P and Q have force constants k_P and k_Q respectively, where $k_P = 2k_Q$. If both springs are stretched by applying equal force F , and W_P and W_Q represent the work done in stretching them, then:

[NCERT Page 80, 81]

- (1) $W_P = 2W_Q$ (2) $W_Q = 2W_P$
 (3) $W_P = 4W_Q$ (4) $W_P = W_Q$

2. A particle moves in a conservative force field where the potential energy is given by $U(x, y, z) = 3x^2y - 4yz + 5z$. The force acting on the particle at point (1, 1, 1) is:

[NCERT Page 78]

- (1) $-6\hat{i} - (3 - 4)5 - (-4 + 5)\hat{k}$
 (2) $-6\hat{i} + 5 - \hat{k}$
 (3) $-6\hat{i} - 5 - \hat{k}$
 (4) $6\hat{i} - 5 + \hat{k}$

3. The acceleration due to gravity on the surface of earth at the equator becomes zero due to rotation of earth about its polar axis. The angular velocity of the earth in that case should be ($g = 10 \text{ m/s}^2$, $R = 6400 \text{ km}$):

[NCERT Page 133, 134]

- (1) $1.25 \times 10^3 \text{ rad/s}$ (2) $1.56 \times 10^3 \text{ rad/s}$
 (3) $1.25 \times 10^2 \text{ rad/s}$ (4) $2.5 \times 10^3 \text{ rad/s}$

4. A non-linear spring exerts a restoring force given by $F = -kx - \frac{1}{2}x^3$ where k and $\frac{1}{2}$ are positive constants. The work done by the agent in stretching the spring slowly from $x = 0$ to $x = L$ is:

[NCERT Page 75]

- (1) $(\frac{1}{2})kL^2 + (\frac{1}{4})^2L^4$
 (2) $kL^2 + \frac{1}{2}L^4$
 (3) $(\frac{1}{2})kL^2 + (\frac{1}{3})^2L^3$
 (4) $(\frac{1}{3})kL^3 + (\frac{1}{4})^2L^4$

5. Three identical particles, each of mass m , are situated at the vertices of an equilateral triangle of side a . Under mutual gravitational attraction, they revolve in a circular orbit circumscribing the triangle. The orbital speed of each particle is:

[NCERT Page 130]

- (1) $\sqrt{\frac{Gm}{a}}$ (2) $\sqrt{\frac{2Gm}{a}}$
 (3) $\sqrt{\frac{Gm}{(\frac{1}{3}a)}}$ (4) $\sqrt{\frac{3Gm}{a}}$

6. A satellite revolves in a circular orbit around a planet of mass M and radius R at an altitude $h = R$ above the surface. If T is the time period of this satellite, then the time period of another satellite orbiting at altitude $h = 7R$ is:

[NCERT Page 130]

- (1) $4T$ (2) $8T$
 (3) $2\sqrt{2}T$ (4) $16T$

7. A vehicle of mass m starts from rest and is driven by an engine delivering constant power P . Neglecting friction, the velocity v attained by the vehicle as a function of distance s travelled is proportional to:

[NCERT Page 83]

- (1) $s^{(1/2)}$ (2) $s^{(1/3)}$
 (3) $s^{(2/3)}$ (4) $s^{(3/2)}$

8. A block of mass m is dropped from a height h above a vertical light spring of spring constant k mounted on a flat horizontal floor. If the maximum compression produced in the spring is x , the equation relating h , x and the parameters is:

[NCERT Page 134]

- (1) $mg(h + x) = (\frac{1}{2})kx^2$
 (2) $mgh = (\frac{1}{2})kx^2$
 (3) $mg(h - x) = (\frac{1}{2})kx^2$
 (4) $mg(h + x) = kx^2$

9. The escape speed v_e from a planet of mass M and radius R is v_e . If the radius of the planet is shrunk by 1% keeping its mass constant, the escape speed will:

[NCERT Page 136]

- (1) increase by 0.5% (2) decrease by 0.5%
 (3) increase by 1% (4) decrease by 2%

10. A ball of mass m moving with velocity u collides head-on elastically with another stationary ball of mass $3m$. The fractional loss of kinetic energy of the incoming ball of mass m is:

[NCERT Page 84, 85]

- (1) $3/4$ (2) $1/4$
(3) $8/9$ (4) $7/16$

11. The density inside a solid sphere of radius R varies with distance r from its centre as $\rho(r) = \rho_0 (r/R)$. The gravitational field inside the sphere at distance r is proportional to:

[NCERT Page 130]

- (1) r (2) r^2
(3) $r^{1/2}$ (4) $1/r$

12. A horizontal turntable rotates about a vertical central axis with angular velocity ω . A child of mass m sits at the centre. When the child walks outward along a radius to the edge, the moment of inertia doubles. The final kinetic energy is:

[NCERT Page 119]

- (1) $E\omega / 2$ (2) $E\omega / 4$
(3) $2E\omega$ (4) $E\omega$

13. The gravitational potential due to a mass distribution along the x -axis is given by $V(x) = -C / (x^2 + a^2)$. The magnitude of the field at distance x from the origin is:

[NCERT Page 134, 135]

- (1) $Cx / (x^2 + a^2)^{3/2}$
(2) $C / (x^2 + a^2)^{3/2}$
(3) $Cx / (x^2 + a^2)$
(4) $C / (x^2 + a^2)^{1/2}$

14. A smooth wire is bent into a vertical circular arc of radius R subtending 90° at centre. A small ring is released from rest from the top end A and slides down along the wire to bottom B. The normal force exerted by the wire on the ring as it reaches B is:

[NCERT Page 78]

- (1) mg (2) $2mg$
(3) $3mg$ (4) $4mg$

15. A thin uniform rod of length L and mass M is placed along the x -axis from $x = d$ to $x = d + L$. The gravitational force exerted by this rod on a point mass m placed at the origin $x = 0$ is:

[NCERT Page 130]

- (1) $GMm / [d(d + L)]$ (2) $GMm / (d + L)^2$
(3) GMm / d^2 (4) $GMm / [2d(d + L)]$

16. A particle of mass m moves in a straight line with velocity v . The kinetic energy is positive and constant. The instantaneous power delivered to the particle by the net force as a function of position x is:

[NCERT Page 83]

- (1) $(1/2) m v^3 x$ (2) $m v^3 x$
(3) $(1/2) m v^2 x$ (4) $m v^2 x$

17. A force $\vec{F} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ N acts on a rigid body at a point $(1, 2, 5)$ m relative to the origin. The torque about the origin is:

[NCERT Page 106]

- (1) $(5\hat{i} + 6\hat{j} + 7\hat{k})$ N·m
(2) $(5\hat{i} - 2\hat{j} + 7\hat{k})$ N·m
(3) $(-5\hat{i} + 6\hat{j} - 7\hat{k})$ N·m
(4) $(7\hat{i} + 6\hat{j} + 7\hat{k})$ N·m

18. Three identical solid spheres of mass M and radius R are arranged such that each touches the other two on a horizontal plane. The moment of inertia of the system about an axis passing through the centroid of the triangle and perpendicular to the plane of centres is:

[NCERT Page 115, 116]

- (1) $(26/5) MR^2$ (2) $(16/5) MR^2$
(3) $(36/5) MR^2$ (4) $(18/5) MR^2$

19. Four particles each of mass M are situated at the four vertices of a square of side a . The gravitational potential at the centre of the square is:

[NCERT Page 130]

- (1) $-4 \frac{GM}{a}$ (2) $-2 \frac{GM}{a}$
(3) $-8 \frac{GM}{a}$ (4) $-4 \frac{GM}{a}$

20. A body of mass m is dropped from a height H above the ground. If air resistance exerts a constant upward opposing force f , its kinetic energy just before striking the ground is:

[NCERT Page 76, 78]

- (1) $mgH - fH$ (2) $mgH + fH$
(3) fH (4) $(mg - f) / H$

21. Point masses $m, 2m, 3m, \dots, nm$ are placed along the x -axis at distances $d, 2d, 3d, \dots, nd$ respectively from the origin. The position of the center of mass of the system is:

[NCERT Page 96]

- (1) $(2n + 1)d / 3$ (2) $(n + 1)d / 2$
(3) $n(n + 1)d / 2$ (4) $2d / (n + 1)$

22. The orbital radius of satellite A is 4 times that of satellite B around the Earth. If the speed of satellite A is v , then the orbital speed of satellite B is:

[NCERT Page 137]

- (1) $2v$ (2) $v / 2$
(3) $4v$ (4) $v / 4$

23. A particle moves in a circular path of radius R . If its kinetic energy K is proportional to s^2 (where s is the distance travelled), the tangential acceleration at is:

[NCERT Page 106]

- (1) proportional to s
(2) constant
(3) proportional to s^2
(4) inversely proportional to s

24. A solid cylinder of mass M and radius R rolls down an inclined plane of inclination θ , without slipping. The acceleration of the cylinder is:

[NCERT Page 120]

- (1) $(2/3) g \sin \theta$ (2) $(1/2) g \sin \theta$
(3) $(3/5) g \sin \theta$ (4) $g \sin \theta$

25. A body of mass 2 kg moves along the x -axis such that its position is given by $x(t) = 2t^3 - 3t^2 + 4$ (in SI units). The net work done on the body from $t = 0$ to $t = 2 \text{ s}$ is:

[NCERT Page 76]

- (1) 144 J (2) 72 J
(3) 288 J (4) 36 J

26. The gravitational force between two masses is $F = -k/r^2$ (where k is a constant). The relation between orbital radius r and orbital period T for circular planetary motion would be:

[NCERT Page 74]

- (1) $T^2 \propto r$ (2) $T^2 \propto r^3$
(3) $T^2 \propto r^3$ (4) $T \propto r^2$

27. A uniform meter stick of mass 150 g is pivoted at the 40 cm mark. What mass must be hung from the 10 cm mark to balance the stick horizontally?

[NCERT Page 109]

- (1) 50 g (2) 75 g
(3) 100 g (4) 150 g

28. A constant net torque of $20 \text{ N}\cdot\text{m}$ acts on a body of moment of inertia $4 \text{ kg}\cdot\text{m}^2$ initially at rest. The angular momentum of the body after 5 seconds is:

[NCERT Page 107]

- (1) $100 \text{ kg}\cdot\text{m}^2/\text{s}$ (2) $50 \text{ kg}\cdot\text{m}^2/\text{s}$
(3) $80 \text{ kg}\cdot\text{m}^2/\text{s}$ (4) $200 \text{ kg}\cdot\text{m}^2/\text{s}$

29. A point mass m is placed on the axis of a uniform ring of mass M and radius R at a distance $x = \sqrt{3} R$ from its center. The gravitational potential at that point is:

[NCERT Page 138]

- (1) $-GM / (2R)$ (2) $-GM / (\sqrt{3} R)$
(3) $-GM / R$ (4) $-2GM / R$

30. A flywheel rotates with a constant angular acceleration α . If it covers 20 radians in the first 2 seconds starting from rest, the angle rotated in the next 2 seconds is:

[NCERT Page 117]

- (1) 60 rad (2) 40 rad
(3) 80 rad (4) 100 rad

31. Three point masses each of mass m are located at the vertices of an equilateral triangle of side L . The work done by an external agent in doubling the separation between each pair of masses is:

[NCERT Page 130]

- (1) $3Gm^2 / (2L)$ (2) $3Gm^2 / L$
(3) $Gm^2 / (2L)$ (4) $6Gm^2 / L$

32. A body of mass m moving with speed v makes a completely inelastic head-on collision with a body of mass $4m$ at rest. The percentage loss in kinetic energy of the system is:

[NCERT Page 84, 85]

- (1) 80% (2) 20%
(3) 75% (4) 50%

33. A particle of mass m moves along the line $y = b$ with constant speed v parallel to the x -axis in the positive x -direction. Its angular momentum about the origin is:

[NCERT Page 106, 107]

- (1) constant and equal to $-m v b k$
(2) increases linearly with time
(3) zero
(4) depends on x

34. Two planets A and B have radii in the ratio $2 : 1$ and densities in the ratio $1 : 2$. The ratio of acceleration due to gravity on their surfaces ($g_A : g_B$) is:

[NCERT Page 129]

- (1) $1 : 1$ (2) $2 : 1$
(3) $1 : 2$ (4) $4 : 1$

35. A bullet of mass 20 g moving with velocity 400 m/s strikes a wooden block of mass 1.98 kg suspended by a light vertical cord of length 1 m and gets embedded in it. The maximum angle through which the string swings is ($g = 10 \text{ m/s}^2$):

[NCERT Page 84]

- (1) 60° (2) 30°
(3) 45° (4) 90°

36. Water is falling from a height of 50 m at a rate of 15 kg/s to operate a turbine. The losses due to frictional forces are 10% of energy. The power generated by the turbine is ($g = 10 \text{ m/s}^2$):

[NCERT Page 83]

- (1) 6.75 kW (2) 7.5 kW
(3) 8.25 kW (4) 7.0 kW

37. A planet moves in an elliptical orbit around the Sun. If the orbital speed at perihelion is v_p and at aphelion is v_a , the ratio of orbital speed at perihelion to that at aphelion is:
- [NCERT Page 128]
- (1) r_p / r_a (2) r_a / r_p
 (3) $(r_p / r_a)^2$ (4) $(r_a / r_p)^2$
38. A solid sphere and a hollow cylinder of equal mass and equal radius roll down an inclined plane from the same height without slipping. Which reaches the bottom first?
- [NCERT Page 120]
- (1) Solid sphere
 (2) Hollow cylinder
 (3) Both reach simultaneously
 (4) Depends on angle of inclination
39. The potential energy of a 1 kg particle moving along x-axis is given by $V(x) = (x^2 - 2x^3) \text{ J}$. The particle is released from rest at $x = 0$. The maximum speed during subsequent motion is:
- [NCERT Page 78]
- (1) 3 m/s (2) 4 m/s
 (3) 2 m/s (4) 7 m/s
40. A force $F = (10 + 0.5x) \text{ N}$ acts on a particle in the x-direction, where x is in meters. The work done by this force during a displacement from $x = 0$ to $x = 2 \text{ m}$ is:
- [NCERT Page 76]
- (1) 21 J (2) 20 J
 (3) 22 J (4) 25 J
41. The radius of gyration of a uniform circular disc of mass M and radius R about a tangent in the plane of the disc is:
- [NCERT Page 114, 115]
- (1) $(\frac{5}{2}) R$ (2) $(\frac{5}{3}) R$
 (3) $(\frac{1}{2}) R$ (4) $(\frac{3}{2}) R$
42. Two discs of moments of inertia I_1 and I_2 rotate with angular velocities ω_1 and $2\omega_1$ respectively in the same sense about a common frictionless axis. When brought into face-to-face contact, the common angular velocity is:
- [NCERT Page 119, 122]
- (1) $(5/3) \omega_1$ (2) $(3/2) \omega_1$
 (3) $2 \omega_1$ (4) $(4/3) \omega_1$
43. The escape velocity of a body from the surface of Earth is 11.2 km/s. If a body is projected with twice the escape speed ($v = 2v_e$), its speed in interstellar space (far away from Earth) will be:
- [NCERT Page 136, 137]
- (1) 11.2 km/s (2) 11.2 km/s
 (3) 22.4 km/s (4) 11.2 km/s
44. A spherical cavity of radius $R/2$ is scooped out of a solid uniform sphere of radius R and mass M, touching the outer surface. The distance of the center of mass of the remaining portion from the center of the original sphere is:
- [NCERT Page 130]
- (1) $R/14$ (2) $R/7$
 (3) $R/6$ (4) $R/8$
45. A block of mass 2 kg is pulled up along an inclined plane of inclination 30° by a force of 20 N parallel to the plane. The coefficient of kinetic friction is 0.2. The work done against gravity when the block moves 5 m along the incline is ($g = 10 \text{ m/s}^2$):
- [NCERT Page 74]
- (1) 50 J (2) 100 J
 (3) 75 J (4) 25 J

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46. The standard enthalpy of formation ($\Delta_f H^\circ$) of $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$ and $\text{CH}_4(\text{g})$ are -393.5 , -285.8 and $-74.8 \text{ kJ mol}^{-1}$ respectively. The standard enthalpy of combustion of methane at 298 K is: [NCERT Page 125]

- [NCERT Page 143]
- (1) $-890.3 \text{ kJ mol}^{-1}$ (2) $-604.5 \text{ kJ mol}^{-1}$
 (3) $-965.1 \text{ kJ mol}^{-1}$ (4) $+890.3 \text{ kJ mol}^{-1}$

47. From the following, select the correct statements regarding SF_6 molecule:

- [NCERT Page 120]
- (1) A, B and C only (2) A and B only
 (3) B, C and D only (4) A, B, C and D

48. Given the bond enthalpies: $\text{C-H} = 414 \text{ kJ mol}^{-1}$, $\text{C-C} = 347 \text{ kJ mol}^{-1}$, $\text{C=C} = 614 \text{ kJ mol}^{-1}$, and $\text{H-H} = 436 \text{ kJ mol}^{-1}$. The enthalpy change for the hydrogenation of ethene: $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$ is: [NCERT Page 153]

- (1) -125 kJ mol^{-1} (2) $+125 \text{ kJ mol}^{-1}$
 (3) -250 kJ mol^{-1} (4) $+250 \text{ kJ mol}^{-1}$

49. According to Fajan's rules, which among the following compounds has the highest covalent character?

- [NCERT Page 112]
- (1) AlCl_3 (2) MgCl_2
 (3) NaCl (4) SiCl_4

50. Two moles of an ideal gas expand isothermally and reversibly from 2 L to 20 L at 300 K. The work done by the gas is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$): [NCERT Page 142]

- (1) -11.49 kJ (2) -5.74 kJ
 (3) $+11.49 \text{ kJ}$ (4) -22.98 kJ

51. Match the species in Column-I with their molecular shapes in Column-II based on VSEPR theory:

- (1) A - (q), B - (p), C - (s), D - (r)
 (2) A - (p), B - (q), C - (r), D - (s)
 (3) A - (q), B - (s), C - (p), D - (r)
 (4) A - (s), B - (p), C - (q), D - (r)

52. Which of the following pairs of molecules has the same bond order?

- [NCERT Page 111]
- (1) O_2 and NO (2) CO and NO_2
 (3) N_2 and O_2 (4) C_2 and N_2

53. Given below are two statements:

- Statement I: The boiling point of H_2O is abnormally high due to extensive intermolecular hydrogen bonding.
 Statement II: Hydrogen bonding is stronger in HF than in HCl because fluorine is more electronegative than oxygen.

- [NCERT Page 131]
- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

54. For the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, if $\Delta_r H^\circ = -92.4 \text{ kJ mol}^{-1}$, then the value of $\Delta_r U^\circ$ at 298 K is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$): [NCERT Page 143]

- (1) -87.45 kJ (2) -97.35 kJ
 (3) -92.40 kJ (4) -82.50 kJ

55. Under which of the following conditions is ΔH_{eq} of a chemical reaction involving gases?

[NCERT Page 143]

- (1) When $\Delta n_g = 0$
- (2) When the reaction is carried out at constant volume
- (3) When the reaction is carried out at constant pressure only
- (4) Both (1) and (2)

56. According to VSEPR theory, the repulsive interactions follow the order:

[NCERT Page 114]

- (1) Lone pair - Lone pair > Lone pair - Bond pair > Bond pair - Bond pair
- (2) Bond pair - Bond pair > Lone pair - Bond pair > Lone pair - Lone pair
- (3) Lone pair - Bond pair > Lone pair - Lone pair > Bond pair - Bond pair
- (4) Lone pair - Lone pair > Bond pair - Bond pair > Lone pair - Bond pair

57. One mole of ice at 0°C melts to liquid water at 0°C . If the molar enthalpy of fusion of ice is 6.0 kJ mol^{-1} , the entropy change is:

[NCERT Page 162]

- (1) $21.98 \text{ J K}^{-1} \text{ mol}^{-1}$
- (2) $60.0 \text{ J K}^{-1} \text{ mol}^{-1}$
- (3) $219.8 \text{ J K}^{-1} \text{ mol}^{-1}$
- (4) $6.0 \text{ J K}^{-1} \text{ mol}^{-1}$

58. Which of the following molecules has zero dipole moment?

[NCERT Page 111]

- (1) BeCl_2 and BCl_3
- (2) H_2O and NH_3
- (3) SO_2 and CHCl_3
- (4) N_2 and CO_2

59. The formal charge on the central oxygen atom in the ozone molecule (O_3) is:

[NCERT Page 102]

- (1) +1
- (2) 0
- (3) -1
- (4) +2

60. Given below are two statements:

Statement I: For an isolated system undergoing an irreversible spontaneous process, the entropy of the system always increases ($\Delta S > 0$).

Statement II: For a cyclic reversible process, the total entropy change of the universe is zero.

[NCERT Page 162]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

61. The hybridization of central atoms in IF_5 , PCl_5 , and SF_6 respectively:

[NCERT Page 120]

- (1) sp^3d , sp^3d , sp^3d^2
- (2) sp^3d , sp^3d^2 , sp^3d
- (3) sp^3d^2 , sp^3d , sp^3d^2
- (4) sp^3 , sp^3d , sp^3d^2

62. For the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H^\circ = -296.7 \text{ kJ mol}^{-1}$. Standard enthalpy of formation of $\text{SO}_3(\text{g})$ is $-395.7 \text{ kJ mol}^{-1}$. Standard enthalpy of formation of $\text{SO}_2(\text{g})$ is:

[NCERT Page 154]

- (1) $-296.7 \text{ kJ mol}^{-1}$
- (2) $-494.7 \text{ kJ mol}^{-1}$
- (3) $+296.7 \text{ kJ mol}^{-1}$
- (4) $-198.0 \text{ kJ mol}^{-1}$

63. Match the species in Column-I with their bond orders in Column-II:

[NCERT Page 125]

- (1) A - (p), B - (q), C - (r), D - (s)
- (2) A - (q), B - (p), C - (s), D - (r)
- (3) A - (s), B - (r), C - (q), D - (p)
- (4) A - (p), B - (r), C - (q), D - (s)

64. Which of the following statements is TRUE regarding an adiabatic reversible expansion of an ideal gas?

[NCERT Page 154]

- (1) $q = 0$, $\Delta U = w < 0$, and temperature drops
 (2) $q > 0$, $\Delta U = 0$, and temperature remains constant
 (3) $q = 0$, $\Delta U = 0$, and temperature rises
 (4) $w = 0$, $q = \Delta U$, and temperature drops

65. The heat of combustion of carbon to CO, is $-393.5 \text{ kJ mol}^{-1}$ and of CO to CO_2 , is $-283.0 \text{ kJ mol}^{-1}$. The enthalpy of formation of CO_2 is:

[NCERT Page 154]

- (1) $-110.5 \text{ kJ mol}^{-1}$ (2) $+110.5 \text{ kJ mol}^{-1}$
 (3) $-676.5 \text{ kJ mol}^{-1}$ (4) $-55.25 \text{ kJ mol}^{-1}$

66. For the reaction: $\text{A(g)} + \text{B(g)} \rightarrow \text{C(g)} + \text{D(g)}$, $\Delta H = -40 \text{ kJ mol}^{-1}$ and $\Delta G = -50 \text{ J K}^{-1} \text{ mol}^{-1}$. The temperature above which the reaction becomes non-spontaneous is:

[NCERT Page 154]

- (1) 800 K (2) 400 K
 (3) 600 K (4) 1200 K

67. Match the state functions in Column-I with their mathematical relationships in Column-II for an ideal gas:

[NCERT Page 143]

- (1) A - (p), B - (q), C - (r), D - (s)
 (2) A - (q), B - (p), C - (s), D - (r)
 (3) A - (r), B - (s), C - (p), D - (q)
 (4) A - (s), B - (p), C - (q), D - (r)

68. For the dissociation reaction: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ at 25°C , $\Delta H^\circ = 124.0 \text{ kJ mol}^{-1}$. The value of ΔU° at 25°C is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$):

[NCERT Page 142]

- (1) $121.52 \text{ kJ mol}^{-1}$ (2) $126.48 \text{ kJ mol}^{-1}$
 (3) $124.0 \text{ kJ mol}^{-1}$ (4) $118.25 \text{ kJ mol}^{-1}$

69. In which of the following pairs of molecules do the central atoms exhibit sp^3d^2 hybridization?

[NCERT Page 120]

- (1) SF_6 and XeF_6 (2) PCl_5 and BrF_5
 (3) IF_7 and SF_6 (4) XeF_4 and IF_7

70. Which of the following sets contains ONLY intensive properties?

[NCERT Page 144]

- (1) Density, temperature, refractive index, molar heat capacity
 (2) Volume, mass, enthalpy, entropy
 (3) Heat capacity, density, pressure, mass
 (4) Internal energy, mole fraction, surface tension, volume

71. Given below are two statements:

Statement I: An endothermic reaction with positive ΔH° can be spontaneous at high temperatures.
 Statement II: The Gibbs free energy change (ΔG) becomes negative when $T\Delta S$ exceeds ΔH .

[NCERT Page 158]

- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

72. For a process occurring at constant temperature and pressure, the criterion for spontaneity in terms of system variables is:

[NCERT Page 157]

- (1) $\Delta G_{\text{sys}} < 0$ (2) $\Delta S_{\text{sys}} > 0$
 (3) $\Delta H_{\text{sys}} < 0$ (4) $\Delta U_{\text{sys}} < 0$

73. According to Molecular Orbital Theory, the total number of anti-bonding electrons in O_2 molecule is:

[NCERT Page 125]

- (1) 6 (2) 4
(3) 8 (4) 2

74. Which of the following diatomic species is diamagnetic and possesses a bond order of 2?

[NCERT Page 125]

- (1) C_2 (2) O_2
(3) B_2 (4) N_2

75. Which of the following species has all paired electrons and is diamagnetic?

[NCERT Page 125]

- (1) N_2 (2) NO
(3) O_2 (4) B_2

76. Which of the following alkali metal halides has the highest lattice energy?

[NCERT Page 112]

- (1) LiF (2) NaCl
(3) KBr (4) CsI

77. Given below are two statements:

Statement I: The bond angle in H_2O (104.5°) is smaller than in NH_3 (107°) because H_2O has two lone pairs on oxygen causing lone pair-lone pair repulsion.

Statement II: Both H_2O and NH_3 have sp^3 hybridized central atoms.

[NCERT Page 112]

- (1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

78. The shape of the BrF_3 molecule is:

[NCERT Page 120]

- (1) Square pyramidal
(2) Trigonal bipyramidal
(3) Pentagonal planar
(4) See-saw

79. Which of the following compounds exhibits both intramolecular and intermolecular hydrogen bonding in different isomers?

[NCERT Page 111]

- (1) o-Nitrophenol and p-Nitrophenol
(2) Water and ammonia
(3) Ethanol and dimethyl ether
(4) o-Cresol and m-Cresol

80. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): CCl_4 is non-polar and insoluble in water. $CHCl_3$ has a net dipole moment.

Reason (R): In CCl_4 , the individual C-Cl bond dipoles cancel out symmetrically due to regular tetrahedral geometry.

[NCERT Page 107, 111]

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is NOT the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) is incorrect but (R) is correct.

81. A reaction has $\Delta H = +30 \text{ kJ mol}^{-1}$ and $\Delta S = +100 \text{ J K}^{-1} \text{ mol}^{-1}$. The minimum temperature above which the reaction becomes spontaneous is:

[NCERT Page 158]

- (1) 300 K (2) 30 K
(3) 3000 K (4) 273 K

82. Which of the following bonds has the highest percent ionic character?

[NCERT Page 112]

- (1) Cs-F (2) Na-Cl
(3) K-Br (4) Li-I

83. An ideal gas expands from 1.0 L to 3.0 L against a constant external pressure of 2.0 atm. During this process, 250 J of heat is absorbed by the gas. The change in internal energy (ΔU) is (1 L · atm = 101.3 J):

[NCERT Page 143]

- (1) -155.2 J (2) +155.2 J
(3) -405.2 J (4) +405.2 J

84. From the given thermochemical data at 298 K:

$C(s, \text{graphite}) + O_2(g) \rightarrow CO_2(g), \Delta H^\circ = -393.5 \text{ kJ mol}^{-1}$

$C(s, \text{diamond}) + O_2(g) \rightarrow CO_2(g), \Delta H^\circ = -395.4 \text{ kJ mol}^{-1}$

The enthalpy change for the transition $C(\text{graphite}) \rightarrow C(\text{diamond})$ is:

[NCERT Page 154]

- (1) +1.9 kJ mol⁻¹ (2) -1.9 kJ mol⁻¹
(3) +788.9 kJ mol⁻¹ (4) -788.9 kJ mol⁻¹

85. According to Molecular Orbital Theory, which of the following molecules has a fractional bond order?

[NCERT Page 125]

- (1) O₂ (2) N₂
(3) C₂ (4) F₂

86. Given below are two statements:

Statement I: The octet rule fails to explain the stability of odd-electron molecules such as NO and NO₂.

Statement II: In PCl₅ and SF₆, the central atom possesses expanded octet with more than 8 valence electrons.

[NCERT Page 105]

- (1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

87. Consider the following statements regarding hydrogen bonding:

A. Hydrogen bond is a weak electrostatic dipole-dipole attraction.

B. Density of ice is less than that of liquid water because ice has an open cage-like structure held by hydrogen bonds.

C. Boiling point of NH₃ is higher than PH₃ due to hydrogen bonding.

The correct statement(s) is/are:

[NCERT Page 131]

- (1) A, B and C (2) A and B only
(3) B and C only (4) A and C only

88. Match the molecules in Column-I with their central atom hybridization in Column-II:

[NCERT Page 120]

(1) A - (p), B - (q), C - (r), D - (s)

(2) A - (q), B - (p), C - (s), D - (r)

(3) A - (p), B - (r), C - (q), D - (s)

(4) A - (s), B - (q), C - (r), D - (p)

89. Which of the following statements regarding thermodynamic quantities is CORRECT?

A. Enthalpy is an extensive state function.

B. Heat (q) and work (w) are state functions.

C. For an exothermic reaction at standard conditions ΔH° is negative.

D. Absolute entropy of a perfectly crystalline substance at 0 K is zero according to the Third Law.

[NCERT Page 114, 163]

(1) A, C and D only

(2) A and B only

(3) B, C and D only

(4) A, B and C only

90. Match the thermodynamic relations in Column-I with the formulas in Column-II:

[NCERT Page 125]

(1) A - (p), B - (q), C - (r), D - (s)

(2) A - (q), B - (p), C - (s), D - (r)

(3) A - (p), B - (r), C - (q), D - (s)

(4) A - (s), B - (q), C - (r), D - (p)

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91. The rough endoplasmic reticulum (RER) is actively involved in the synthesis of:

[NCERT Page 95]

- (1) Proteins and glycoproteins
- (2) Steroidal hormones and lipids
- (3) Glycogen and cholesterol
- (4) Phospholipids only

92. Identify the statement that is NOT correct regarding leaf anatomy and morphology:

[NCERT Page 60]

- (1) Monocot leaves typically exhibit reticulate venation.
- (2) A typical leaf consists of three main parts: leaf base, petiole, and lamina.
- (3) In monocotyledons, the leaf base expands into a sheath covering the stem partially or wholly.
- (4) Leaves originate from shoot apical meristems and are arranged in an acropetal order.

93. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Mycoplasmas can survive without oxygen and are resistant to penicillin.

Reason (R): Mycoplasmas completely lack a cell wall.

[NCERT Page 89]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

94. In a typical dicotyledonous seed, the protective outer covering consists of an outer testa and an inner:

[NCERT Page 66]

- | | |
|---------------|--------------------|
| (1) Tegmen | (2) Aleurone layer |
| (3) Perisperm | (4) Scutellum |

95. The arrangement of ovules within the ovary is known as placentation. In pea plants, the placentation is:

[NCERT Page 65]

- | | |
|--------------|------------------|
| (1) Marginal | (2) Axile |
| (3) Parietal | (4) Free-central |

96. Match the plant anatomical structures in List-I with their functions/features in List-II:

[NCERT Page 71, 77]

- | |
|------------------------------------|
| (1) A - III, B - IV, C - II, D - I |
| (2) A - II, B - III, C - I, D - IV |
| (3) A - IV, B - I, C - II, D - III |
| (4) A - III, B - I, C - IV, D - II |

97. Given below are two statements:

Statement I: The primary cell wall of a young plant cell is capable of growth, which gradually diminishes as the cell matures.

Statement II: The secondary cell wall is formed on the inner (towards membrane) side of the cell.

[NCERT Page 96, 97]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

98. Which of the following fruits develops from a monocarpellary superior ovary and is one-seeded?

[NCERT Page 66]

- (1) Drupe (Mango)
- (2) Pome (Apple)
- (3) Berry (Tomato)
- (4) Hesperidium (Orange)

99. The 70S ribosomes found in prokaryotes and eukaryotic organelles consist of two subunits:

[NCERT Page 98]

- | | |
|-----------------|-----------------|
| (1) 50S and 30S | (2) 60S and 40S |
| (3) 50S and 40S | (4) 60S and 30S |

100. Which of the following organelle pairs possesses its own DNA, 70S ribosomes, and replicates semi-autonomously?

[NCERT Page 100]

- (1) Mitochondria and Chloroplasts
- (2) Golgi apparatus and Endoplasmic Reticulum
- (3) Lysosomes and Peroxisomes
- (4) Nucleolus and Centrosome

101. Match the cellular structures in List-I with their characteristics in List-II:

[NCERT Page 95, 97-99]

- (1) A - III, B - II, C - I, D - IV
- (2) A - II, B - III, C - I, D - IV
- (3) A - III, B - I, C - II, D - IV
- (4) A - I, B - II, C - III, D - IV

102. Which of the following cellular components is absent in all animal cells but universally present in higher plant cells?

[NCERT Page 94]

- (1) Cellulosic cell wall
- (2) Mitochondria
- (3) Endoplasmic reticulum
- (4) Ribosomes

103. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Lipids are not strictly macromolecules.

Reason (R): Molecular weight of lipids does not exceed 800 Da, but they appear in the acid-insoluble fraction because they form vesicular structures.

[NCERT Page 108, 109]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

104. Consider the following statements regarding the fluid mosaic model of cell membrane:

- A. Integral proteins are partially or totally buried in the membrane.
- B. The quasi-fluid nature of lipid enables lateral movement of proteins within overall bilayer.
- C. Lipids are arranged in a bilayer with polar heads facing the inner aqueous side.

The correct statements are:

[NCERT Page 97]

- (1) A and B only
- (2) B and C only
- (3) A and C only
- (4) A, B and C

105. When a flower can be divided into two similar halves only in one particular vertical plane, it is termed as:

[NCERT Page 62]

- (1) Zygomorphic
- (2) Actinomorphic
- (3) Asymmetric
- (4) Epigynous

106. Which of the following organelles is NOT considered a part of the endomembrane system?

[NCERT Page 95]

- (1) Peroxisome and Mitochondria
- (2) Endoplasmic Reticulum
- (3) Golgi complex
- (4) Lysosome

107. From the statements given below, choose the CORRECT option regarding chloroplasts:

- A. Chloroplasts are double-membrane bound organelles.
- B. The stroma contains enzymes required for the synthesis of carbohydrates and proteins.
- C. Thylakoids are arranged in stacks like the piles of coins called grana.
- D. Inner membrane of chloroplast is more permeable than the outer membrane.

[NCERT Page 96, 97]

- (1) A, B and C only
- (2) B, C and D only
- (3) A and D only
- (4) A, B, C and D

108. Which of the following conditions is observed when stamens are attached to petals?

[NCERT Page 64]

- (1) Epipetalous (e.g., Brinjal)
- (2) Epiphyllous (e.g., Lily)
- (3) Monadelphous (e.g., China rose)
- (4) Diadelphous (e.g., Pea)

109. Starch gives an intense blue-black colour with iodine solution because:

[NCERT Page 110]

- (1) Starch forms helical secondary structures that trap I₂ in the interior of helical coils.
- (2) Starch chemically oxidizes iodine into iodide ions.
- (3) Cellulose and starch both have unbranched linear helical chains.
- (4) Starch has free reducing aldehyde groups that react with iodine.

110. The major site for the synthesis of steroidal hormones and lipids in animal cells is:

[NCERT Page 97]

- (1) Smooth Endoplasmic Reticulum (SER)
- (2) Rough Endoplasmic Reticulum (RER)
- (3) Golgi apparatus
- (4) Peroxisome

111. In which of the following plants is the edible part correctly matched?

[NCERT Page 66]

- (1) Coconut - Endosperm
- (2) Mango - Endocarp
- (3) Apple - Mesocarp only
- (4) Tomato - Pericarp only

112. In ribosomes, the letter "S" stands for the Svedberg unit, which is an indirect measure of:

[NCERT Page 98]

- (1) Density and size (Sedimentation coefficient)
- (2) Weight and volume
- (3) Number of RNA molecules
- (4) Electrical charge

113. Given below are two statements:

Statement I: Epidermal hairs on the stem called trichomes are usually multicellular and prevent water loss due to transpiration.

Statement II: Root hairs are unicellular elongations of epidermal cells that absorb water and minerals from the soil.

[NCERT Page 89]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

114. Match the chromosome terms in List-I with their descriptions in List-II:

[NCERT Page 99, 101, 102]

- (1) A - II, B - I, C - III, D - IV
- (2) A - I, B - II, C - III, D - IV
- (3) A - II, B - III, C - I, D - IV
- (4) A - III, B - I, C - II, D - IV

115. Which of the following seeds is non-endospermic (exalbuminous)?

[NCERT Page 66, 67]

- | | |
|------------------|------------|
| (1) Gram and Pea | (2) Castor |
| (3) Maize | (4) Wheat |

116. The sequential pathway for protein secretion from a eukaryotic cell is:

[NCERT Page 95, 96]

- (1) Rough ER !' Transport vesicle !' Cis face of Golgi !' Secretory vesicle !' Plasma membrane
- (2) Rough ER !' Lysosome !' Trans face of Golgi !' Secretory vesicle !' Plasma membrane
- (3) Smooth ER !' Rough ER !' Golgi cisternae !' Nuclear envelope !' Plasma membrane
- (4) Golgi trans face !' Cis face !' RER !' Plasma membrane

117. In a dorsiventral (dicotyledonous) leaf, which of the following is correct?

[NCERT Page 76]

- (1) Stomata are generally more numerous on the abaxial (lower) epidermis.
- (2) Mesophyll is not differentiated into palisade and spongy tissue.
- (3) Stomata are equally distributed on both surfaces (amphistomatic).
- (4) Vascular bundles are all of identical size irrespective of venation.

118. In coconut fruit, the fibrous portion used for coir manufacture is derived from:

[NCERT Page 65, 66]

- | | |
|--------------|--------------|
| (1) Mesocarp | (2) Endocarp |
| (3) Epicarp | (4) Testa |

119. Choose the CORRECT statement regarding eukaryotic cilia and flagella:

- A. Core of cilium is called axoneme.
- B. Axoneme usually has nine pairs of doublets of radially arranged peripheral microtubules and a pair of central microtubules (9 + 2 array).
- C. Both cilium and flagellum emerge from centriole-like structures called basal bodies.
- D. Peripheral doublets are interconnected by radial spokes.

[NCERT Page 99]

- | | |
|-------------------|---------------------|
| (1) A, B, C and D | (2) A, B and C only |
| (3) B and D only | (4) A and C only |

120. Match the types of placentation in List-I with their representative plants in List-II:

[NCERT Page 65]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

- 121.** Choose the correct statements regarding bacterial cell envelope:
 A. It consists of a tightly bound three-layered structure: glycocalyx, cell wall, and plasma membrane.
 B. Glycocalyx could be a loose sheath called slime layer or thick and tough called capsule.
 C. Mesosomes are formed by invaginations of plasma membrane into the cell.
 D. Gram-positive bacteria have thick lipopolysaccharide outer membranes.

[NCERT Page 90, 91]

- (1) A, B and C only (2) B, C and D only
 (3) A and D only (4) A, B, C and D

- 122.** A chromosome having a terminal centromere is called:

[NCERT Page 101]

- (1) Telocentric (2) Metacentric
 (3) Sub-metacentric (4) Acrocentric

- 123.** Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Vascular bundles in monocot stem are conjoint, collateral, and closed.

Reason (R): Cambium is absent between xylem and phloem in monocot vascular bundles.

[NCERT Page 73]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.

- 124.** In a monocot seed embryo (e.g., maize), the shield-shaped single large cotyledon is known as:

[NCERT Page 67]

- (1) Scutellum (2) Coleoptile
 (3) Coleorhiza (4) Epiblast

- 125.** When stamens are united into two bundles, the condition is termed:

[NCERT Page 64]

- (1) Diadelphous (as in Pea)
 (2) Monadelphous (as in China rose)
 (3) Polyadelphous (as in Citrus)
 (4) Syngenesious (as in Sunflower)

- 126.** The primary root and its branches constitute the tap root system. It originates directly from the elongation of:

[NCERT Page 68]

- (1) Radicle (2) Plumule
 (3) Hypocotyl (4) Cotyledonary node

- 127.** Match the cell organelles in List-I with their principal functions in List-II:

[NCERT Page 95-98]

- (1) A - I, B - II, C - III, D - IV
 (2) A - II, B - I, C - IV, D - III
 (3) A - I, B - III, C - II, D - IV
 (4) A - IV, B - II, C - I, D - III

- 128.** Match the vascular bundle arrangements in List-I with their plant organs in List-II:

[NCERT Page 73]

- (1) A - I, B - II, C - III, D - IV
 (2) A - II, B - I, C - IV, D - III
 (3) A - III, B - II, C - I, D - IV
 (4) A - I, B - III, C - II, D - IV

- 129.** Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Interfascicular cambium and cork cambium are secondary meristems.

Reason (R): They arise from permanent tissues through the process of dedifferentiation.

[NCERT Page 101]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.

- 130.** In a transverse section of a typical dicot root, the innermost layer of the cortex characterized by suberin deposition (Casparian strips) is called:

[NCERT Page 74]

- (1) Endodermis (2) Pericycle
 (3) Epiblema (4) Exodermis

131. Match the stamen conditions in List-I with their botanical examples in List-II:

[NCERT Page 64]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - IV, C - II, D - III
- (4) A - III, B - II, C - I, D - IV

132. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In hypogynous flowers, the ovary is superior.
Reason (R): In hypogynous flowers, the gynoecium occupies the highest position while the other floral whorls are situated below it.

[NCERT Page 62]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

133. Which of the following cell organelles is bounded by a single unit membrane?

[NCERT Page 95]

- (1) Lysosome and Vacuole
- (2) Chloroplast
- (3) Mitochondria
- (4) Nucleus

134. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Cellulose does not give a blue colour with iodine.

Reason (R): Cellulose consists of unbranched β -glucose chains and does not form complex helical structures to hold iodine molecules.

[NCERT Page 110]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

135. Exarch xylem development, where protoxylem lies towards the periphery and metaxylem towards the center, is characteristic of:

[NCERT Page 76]

- (1) Roots (both dicot and monocot)
- (2) Stems (both dicot and monocot)
- (3) Dicot stem only
- (4) Monocot stem only

136. From the statements given below regarding carbohydrates, choose the CORRECT option:

- A. Glycogen is a branched storage polysaccharide in animals and fungi.
- B. Chitin is a complex homopolymer of N-acetylglucosamine (NAG).
- C. Inulin is a polymer of fructose.
- D. Starch is a polymer of galactose.

[NCERT Page 110, 111]

- (1) A, B and C only
- (2) B, C and D only
- (3) A and D only
- (4) A, B, C and D

137. Which of the following is a pyrimidine nitrogenous base found only in RNA and not in DNA?

[NCERT Page 106]

- (1) Uracil
- (2) Thymine
- (3) Cytosine
- (4) Adenine

138. In a polypeptide chain, the first amino acid synthesized at the beginning of the chain corresponds to the:

[NCERT Page 112]

- (1) N-terminal amino acid
- (2) C-terminal amino acid
- (3) Carboxyl terminal
- (4) Phosphorylated residue

139. Inhibition of succinate dehydrogenase by malonate is a classic textbook example of:

[NCERT Page 117]

- (1) Competitive reversible inhibition
- (2) Non-competitive irreversible inhibition
- (3) Allosteric feedback inhibition
- (4) Uncompetitive inhibition

140. Identify the statement that is NOT correct regarding protein structure:

[NCERT Page 111, 112]

- (1) Tertiary structure is completely destroyed upon protein synthesis and is absent in active enzymes.
- (2) Primary structure gives the positional sequence of amino acids.
- (3) Secondary structure includes α -helix and β -pleated sheet.
- (4) Quaternary structure is exhibited by proteins consisting of two or more polypeptide subunits (e.g., hemoglobin).

141. A nucleoside differs from a nucleotide in lacking a:

[NCERT Page 106]

- (1) Phosphate group
- (2) Nitrogenous base
- (3) Ribose/deoxyribose sugar
- (4) Glycosidic bond

142. Match the biomolecules in List-I with their descriptions in List-II:

[NCERT Page 106]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - III, D - IV
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

143. From the statements given below regarding nucleic acids, choose the CORRECT option:

- A. Adenine and Guanine are substituted purines.
- B. Cytosine, Uracil and Thymine are pyrimidines.
- C. Adjacent nucleotides in a polynucleotide chain are linked via 3'-5' phosphodiester bonds.
- D. B-DNA has 10 base pairs per turn with a pitch of 3.4 nm.

[NCERT Page 111]

- (1) A, B, C and D
- (2) A and B only
- (3) A, C and D only
- (4) B and D only

144. Match the nucleotide/nucleoside names in List-I with their categories in List-II:

[NCERT Page 106]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - III, D - IV
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - I, C - II, D - IV

145. Match the cellular components in List-I with their average percentage of total cellular mass in List-II:

[NCERT Page 109]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

146. Match the levels of protein structure in List-I with their features in List-II:

[NCERT Page 112]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

147. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Enzymes increase the rate of biochemical reactions immensely.

Reason (R): Enzymes lower the activation energy required to convert substrate into transition state.

[NCERT Page 112]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

148. Match the secondary metabolites in List-I with their chemical categories in List-II:

[NCERT Page 108]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

149. Match the polysaccharides in List-I with their structural characteristics in List-II:

[NCERT Page 110, 111]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

150. Which of the following is the most abundant protein in the whole of the biosphere?

[NCERT Page 106, 111]

- (1) RuBisCO
- (2) Collagen
- (3) Insulin
- (4) Hemoglobin

151. Match the amino acids in List-I with their functional chemical natures in List-II:

[NCERT Page 106, 108]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

152. Select the INCORRECT match among secondary metabolites and their classes:

[NCERT Page 108]

- (1) Monoterpenes - Curcumin
- (2) Essential oil - Lemon grass oil
- (3) Polymeric substances - Rubber, Gums
- (4) Alkaloids - Morphine, Codeine

153. Match the enzyme categories in List-I with their catalyzed reactions in List-II:

[NCERT Page 106]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

154. Which of the following elements is present in maximum percentage (by weight) in human body?

[NCERT Page 109]

- | | |
|---------------------|---------------------|
| (1) Oxygen (65.0%) | (2) Carbon (18.5%) |
| (3) Nitrogen (3.3%) | (4) Hydrogen (0.5%) |

155. Enzymes generally have maximum catalytic activity at:

[NCERT Page 112, 113]

- (1) Optimum pH and optimum temperature
- (2) 0°C
- (3) 100°C
- (4) Strongly alkaline pH 14

156. In a branched polysaccharide such as glycogen, the linear chain linkages and branch point linkages are respectively:

[NCERT Page 110]

- (1) $\pm$ -1,4-glycosidic and $\pm$ -1,6-glycosidic
- (2) 2 -1,4-glycosidic and 2 -1,6-glycosidic
- (3) $\pm$ -1,6-glycosidic and $\pm$ -1,4-glycosidic
- (4) 2 -1,3-glycosidic and $\pm$ -1,4-glycosidic

157. Identify the statement that is NOT correct regarding biomolecules:

[NCERT Page 110, 111]

- (1) Cellulose can be easily digested by humans as we produce cellulase.
- (2) Glycogen is a branched homopolysaccharide stored in liver and muscle.
- (3) Chitin forms the exoskeleton of arthropods.
- (4) Starch consists of amylose and amylopectin.

158. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Collagen is the most abundant protein in the animal world.

Reason (R): It provides high tensile strength to connective tissues such as tendons and skin.

[NCERT Page 108]

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

159. Identify the statement that is NOT correct regarding cofactors of enzymes:

[NCERT Page 114, 118]

- (1) Cofactors always remain irreversibly bound to the apoenzyme and cannot be separated without denaturing the protein.
- (2) A coenzyme is an organic non-protein compound transiently associated during catalysis.
- (3) A prosthetic group is an organic cofactor tightly bound to the apoenzyme.
- (4) Metal ions form coordination bonds with side chains at the active site.

160. Match the cofactors in List-I with their associated enzymes in List-II:

[NCERT Page 108]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

161. Match the enzyme classes in List-I with their catalyzed reactions in List-II:

[NCERT Page 117]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

162. Given below are two statements:

Statement I: The acid-soluble pool consists of biomicromolecules with molecular weights roughly from 18 to 800 Daltons.

Statement II: The acid-insoluble fraction contains four types of organic compounds: proteins, nucleic acids, polysaccharides, and lipids.

[NCERT Page 108, 109]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

163. In an energy profile diagram of an enzymatic reaction, the presence of an enzyme:

[NCERT Page 115]

- (1) Decreases the activation energy of the reaction
- (2) Increases the activation energy of the reaction
- (3) Changes the overall free energy change (ΔG) of the reaction
- (4) Converts an endergonic reaction into an exergonic reaction

164. Match the enzyme terms in List-I with their definitions in List-II:

[NCERT Page 117]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

165. Match the polymers in List-I with their monomers in List-II:

[NCERT Page 109-111]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

166. Match the plant anatomical terms in List-I with their descriptions in List-II:

[NCERT Page 110]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

167. Given below are two statements:

Statement I: Adult human haemoglobin consists of 4 subunits: two α -type and two β -type chains.

Statement II: This represents a quaternary structure of protein.

[NCERT Page 111, 112]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

168. Identify the statement that is NOT correct regarding secondary metabolites:

[NCERT Page 118]

- (1) All secondary metabolites have direct, identifiable roles in primary physiological processes like photosynthesis and respiration of the producing plant.
- (2) Many secondary metabolites are useful to human welfare (e.g., rubber, drugs, scents, pigments).
- (3) Some secondary metabolites have high ecological importance in defense against herbivores.
- (4) Alkaloids, flavonoids, and essential oils are examples of secondary metabolites.

169. Select the CORRECTLY matched pair:

[NCERT Page 118]

- (1) Catalase - Haem prosthetic group
- (2) Carboxypeptidase - Magnesium cofactor
- (3) Peroxidase - Zinc cofactor
- (4) Apoenzyme - Non-protein part

170. From the statements given below, choose the correct statements:

- A. Oils have lower melting point than fats and remain as liquid in winters.
- B. Palmitic acid has 20 carbon atoms including carboxyl carbon.
- C. Lipids are generally water-insoluble.
- D. Phospholipids have a hydrophilic head and hydrophobic hydrocarbon tails.

[NCERT Page 105, 106]

- (1) A, C and D only
- (2) A and B only
- (3) B, C and D only
- (4) A, B, C and D

171. Given below are two statements:

Statement I: Starch is a homopolymer of glucose and serves as the major energy storage in plants.

Statement II: Glycogen is also a homopolymer of glucose and serves as the storage carbohydrate in animals.

[NCERT Page 110]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

172. Match the steps in catalytic cycle of an enzyme in List-I with their descriptions in List-II:

[NCERT Page 115]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - III, B - II, C - I, D - IV

173. From the statements given below, choose the CORRECT statement(s):

A. In a polysaccharide chain, the right end is called the reducing end and the left end is called the non-reducing end.

B. Cellulose consists of β -glucose units linked by $1 \rightarrow 4$ glycosidic bonds.

C. Starch reacts with iodine to produce a yellow colour.

[NCERT Page 110, 111]

- (1) A and B only
- (2) B and C only
- (3) A and C only
- (4) A, B and C

174. Nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin:

[NCERT Page 117, 118]

- (1) Niacin (Vitamin B₃)
- (2) Thiamine (Vitamin B₁)
- (3) Riboflavin (Vitamin B₂)
- (4) Biotin

175. Given below are two statements:

Statement I: Ribozymes are catalytic RNA molecules that act as biological enzymes.

Statement II: Almost all enzymes are proteins, but some nucleic acids behave like enzymes.

[NCERT Page 106]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

176. Given below are two statements:

Statement I: Primary structure of a protein determines its 3D conformation and functional properties.

Statement II: Peptide bond formation involves a dehydration condensation reaction between $-\text{COOH}$ of one amino acid and the next.

[NCERT Page 111]

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

177. Identify the statement that is NOT correct regarding enzymes:

[NCERT Page 110, 112]

- (1) Competitive inhibitor increases both K_m and V_{max} of the enzymatic reaction.
- (2) Substrate binds specifically at the active site of the enzyme.
- (3) Enzymes work by lowering the activation energy barrier.
- (4) At high temperatures, enzymes lose activity due to thermal denaturation.

178. The graph represents the effect of substrate concentration $[S]$ on the velocity of an enzyme-catalyzed reaction (V) . In the Michaelis-Menten plot, K_m represents:

[NCERT Page 116]

- (1) The substrate concentration at which reaction velocity reaches half of V_{max} ($V_{max} / 2$)
- (2) The velocity at half substrate concentration
- (3) The maximum velocity attained by the enzyme at infinite substrate
- (4) The concentration of enzyme molecules in the solution

179. Match the cofactors in List-I with their chemical nature in List-II:

[NCERT Page 118]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III

180. Match the chemical elements in List-I with their approximate percentage of total human body weight in List-II:

[NCERT Page 105, 109]

- (1) A - I, B - II, C - III, D - IV
- (2) A - II, B - I, C - IV, D - III
- (3) A - I, B - III, C - II, D - IV
- (4) A - IV, B - II, C - I, D - III